

ENTERPRISE CONTEXT BRAIN

Governed Organizational Memory & Agentic Decision Intelligence Platform

AI Hackathon – Product Requirements Document
Version 1.0 | August 2026

Document Item	Details
Product	Enterprise Context Brain
Category	Enterprise AI / Agentic AI / Organizational Memory
Primary Users	Managers, Project Leads, Engineering & Data Teams
Primary Goal	Turn fragmented enterprise information into trusted, actionable context for AI agents
Hackathon Focus	Working MVP demonstrating context-aware managerial decision support

1. Executive Summary

Enterprise Context Brain is a governed organizational memory and agentic decision intelligence platform. It connects fragmented enterprise knowledge—such as project documents, Jira tickets, Git activity, meeting notes, incidents, decisions and operational data—and converts it into trusted, time-aware, evidence-backed context that AI agents can use.

The platform is designed around a key principle: an enterprise AI agent should not operate only from a prompt, a vector search result, or its own conversation history. It should operate from a shared organizational context layer containing facts, decisions, reasons, risks, procedures, outcomes and evidence.

For the hackathon, the MVP will demonstrate a Manager Agent that can answer questions such as: "Why is this project delayed?", "What changed this week?", "What are the biggest current risks?", "Why was this decision made?" and "What action should I take?" with evidence and confidence.

2. Problem Statement

Managers and project teams work across multiple disconnected systems. Important organizational context is distributed across tickets, documents, chat conversations, meeting transcripts, code repositories, dashboards and operational systems.

- Information exists, but the context behind the information is difficult to retrieve.
- Important decisions and the reasons behind them are often buried in meetings, tickets or conversations.
- Current project state may conflict across different systems.
- Traditional RAG can retrieve relevant text but does not inherently understand decision history, freshness, authority or organizational relationships.
- Managers spend significant time manually collecting information before making decisions.
- AI agents may hallucinate, use stale information, access unauthorized data, or perform actions without appropriate approval.

3. Vision

Build a shared, governed organizational context layer that allows enterprise AI agents to understand what the organization knows, why decisions were made, what is currently happening, what has changed, what risks exist, and what actions are permitted.

Core concept:

Traditional AI Assistant	Enterprise Context Brain
Question → Search → Answer	Question → Intent → Context Planning → Memory → Evidence → Reasoning → Action
Mostly document-centric	Decision-, project-, event- and evidence-centric
Vector retrieval	Semantic + structured + graph + temporal retrieval
Agent-specific context	Shared governed organizational memory
Answer-focused	Decision and action focused

4. Target Users

User	Primary Need
Project Manager / Manager	Understand status, risks, changes, decisions and

	recommended actions
Engineering Lead	Understand technical decisions, dependencies, incidents and delivery risks
Data/AI Lead	Understand pipeline, platform, model and project context
Operations Lead	Understand incidents, SLA risks and recurring operational problems
Executive / Leadership	Get concise, evidence-backed organizational insights

5. Primary Use Cases

Use Case	Example Question
Project Health	What is the current state of Project X?
Root Cause	Why is Project X delayed?
Decision Intelligence	Why did we choose this architecture?
Change Detection	What changed in Project X this week?
Risk Intelligence	What are the top risks and who owns them?
Historical Learning	Have we faced a similar issue before?
Action Recommendation	What should I do next?
Proactive Briefing	What should I know today?

6. Proposed Solution

The solution consists of five major capabilities: enterprise ingestion, organizational memory, context intelligence, agent orchestration, and governed action execution.

- Ingest information from enterprise systems.
- Normalize information into canonical organizational entities.
- Store semantic, episodic, procedural, decision and experiential memory.
- Build context dynamically based on user intent, role, freshness and evidence.
- Allow specialized agents to reason over the shared context.
- Expose enterprise tools through controlled tool/MCP interfaces.
- Require human approval for high-impact actions.
- Record every important retrieval, decision, tool call and action for auditability.

7. High-Level Architecture

The architecture separates the organizational memory/data plane from the agent control plane. This allows multiple agents to consume the same governed context without creating isolated knowledge silos.

Manager / User



Manager Context UI



API / Agent Gateway

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Agent Orchestrator

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Context Intelligence Layer

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Organizational Memory

├— Semantic Memory

├— Episodic Memory

├— Procedural Memory

├— Decision Memory

└— Experiential Memory

↓

Structured Store + Vector Index + Optional Knowledge Graph + Object Storage

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Enterprise Connectors / MCP Gateway

├— Jira

├— Git

├— Documents

├— Teams / Collaboration

├— AWS / Cloud

└— Analytics / Operational Systems

8. Organizational Memory Model

Memory Type	Purpose	Example
Semantic	Stable organizational facts	Project uses Databricks
Episodic	Events and history	Deployment failed on Aug 11
Procedural	How work should be performed	Production deployment procedure
Decision	What was decided and why	Architecture selected because of cost and scale
Experiential	Lessons learned from outcomes	Previous migration failed due to missing validation

Core entities:

- Organization, Department, Team, Person
- Project, Product, Customer, Goal
- Decision, Reason, Alternative, Outcome
- Risk, Issue, Incident, Dependency
- Meeting, Policy, Procedure, Event, Lesson
- Evidence, Metric, Agent, Action, Approval

9. Decision Memory

Decision memory is a key differentiator of the platform. A decision is stored as a structured object rather than only as a document chunk.

Field	Example
Decision ID	DEC-2026-0142
Project	KCF
Decision	Move pipeline from X to Y
Owner	Engineering Lead
Reason	Cost and scalability
Alternatives	A, B, C
Evidence	Jira ticket, ADR, meeting
Expected Outcome	30% lower processing cost
Actual Outcome	Pending
Status	Active
Confidence	0.93
Supersedes	DEC-2026-0101

10. Context Intelligence Layer

The Context Intelligence Layer is responsible for selecting the right context instead of simply retrieving the nearest text chunks.

- Intent recognition
- Entity identification
- Context planning
- Hybrid retrieval: semantic + structured + graph
- Role and permission filtering
- Freshness validation
- Source authority ranking
- Contradiction detection
- Context summarization and token optimization
- Evidence attachment
- Confidence scoring

Example flow:

Manager question → Intent = Root Cause Analysis → Identify Project → Retrieve project state, Jira, incidents, decisions, meetings and dependencies → Check freshness and conflicts → Rank evidence → Assemble context → Agent reasoning → Evidence-backed response.

11. Source Trust & Evidence Model

The system must recognize that enterprise sources can disagree. It should rank evidence according to source authority, freshness and verification.

Source Type	Typical Trust Level	Example
System of Record	Very High	Production telemetry / approved system status
Approved Decision	Very High	Architecture Decision Record
Official Ticket	High	Jira issue
Repository	High	Git commit / release
Meeting Notes	Medium	Approved meeting summary
Chat	Medium/Low	Teams message
Agent Inference	Inference	Generated conclusion

Important answers should include evidence, source, timestamp and confidence. When sources conflict, the system should explicitly report the conflict instead of silently choosing one.

12. Agentic Framework

Agent	Responsibility	Example Action
Manager Agent	Orchestrates management questions	Generate project briefing
Project Agent	Project state and delivery context	Summarize project health
Risk Agent	Identify and rank risks	Create risk escalation
Decision Agent	Explain decisions and alternatives	Find decision evidence
Meeting Agent	Extract decisions/actions from meetings	Update organizational memory
Incident Agent	Analyze operational incidents	Find similar incidents

All agents should consume the same governed organizational memory. Agents should not maintain disconnected copies of enterprise knowledge.

13. MCP & Tool Integration

Model Context Protocol (MCP) can provide a standardized interface between agents and enterprise tools. A controlled MCP/tool gateway should enforce authentication, authorization, logging and action policies.

- Jira: read/create/update issues
- Git: retrieve commits, branches, pull requests and release evidence
- Documents: retrieve policies, ADRs and project documents
- Collaboration: retrieve approved meeting context
- AWS/Cloud: retrieve operational status and permitted metadata
- Analytics: retrieve approved metrics and dashboards

14. Governance, Security & Human-in-the-Loop

- Role-based and attribute-based access control
- Source-level permission enforcement
- Agent identity and tool permissions

- Data classification and sensitive-data controls
- Immutable audit trail for important agent actions
- Human approval for high-impact actions
- Ability to correct, supersede or expire memory
- Retention and deletion workflows
- Prompt injection and untrusted-content defenses
- Full execution tracing for agent decisions and tool calls

Risk Level	Example	Execution Policy
Low	Summarize a meeting	Automatic
Medium	Create a Jira issue	Automatic with notification / policy dependent
High	Change production configuration	Human approval required
Critical	Delete data / approve sensitive transaction	Human-only

15. SLA & Quality Model

Initial hackathon targets; these should be validated against the chosen infrastructure before production adoption.

Category	Target
Availability	99.9% for Context API
Retrieval latency	P95 < 500 ms
Critical-source freshness	< 5 minutes where real-time integration exists
Evidence-backed answers	> 95%
Unauthorized access	0
Critical action without approval	0
Agent execution success	> 98% for supported workflows
Critical hallucination	Target 0; investigate every occurrence
Audit coverage	100% of agent actions

16. Hackathon MVP Scope

The MVP should remain intentionally narrow so the team can demonstrate a complete end-to-end agentic workflow.

- One representative project or synthetic enterprise dataset
- Three initial sources: Jira-like tickets, project documents/ADRs and Git activity
- PostgreSQL + pgvector as the initial memory store
- Object storage for source documents
- FastAPI backend
- One Manager Agent plus Project/Risk/Decision capabilities
- Context Intelligence Layer with hybrid retrieval and evidence ranking
- MCP-based tool access for at least one actionable system
- Simple manager dashboard plus conversational interface
- Human approval flow for one high-impact action
- Agent trace and evaluation dashboard

17. Primary Demo Scenario

A manager asks: "Why is Project X delayed, what changed this week, and what should I do?"

1. Manager Agent identifies the intent as project root-cause and action analysis.
2. Context Engine retrieves project status, tickets, incidents, decisions, meetings and dependencies.
3. Memory layer identifies recent changes and historical context.
4. Evidence engine ranks authoritative and fresh sources.
5. Risk Agent identifies the top contributors to the delay.
6. Decision Agent explains any architecture or scope changes.
7. Manager Agent produces a concise answer with evidence and confidence.
8. Agent recommends actions and asks for approval where required.
9. Approved action is executed through the MCP/tool gateway.
10. Execution and outcome are recorded back into organizational memory.

18. Example User Experience

Manager: What are the biggest risks for Project X this week?

Agent response:

Risk	Probability	Impact	Evidence	Recommended Action
AWS access dependency	High	High	Jira + meeting	Escalate dependency
Data pipeline instability	Medium	High	Incident + telemetry	Prioritize validation
Customer approval delay	Medium	Medium	Meeting + ticket	Follow up with owner

Manager: Create an escalation for the AWS dependency.

Agent: This action requires approval. Proceed?

Manager: Approve.

Agent: Escalation created. Action ID and source evidence recorded in the audit trail.

19. Technical Stack – Proposed

Layer	Hackathon Technology
Frontend	React / Next.js
API	Python + FastAPI
Agent orchestration	LangGraph or Microsoft Agent Framework
LLM	Enterprise-approved LLM / model gateway
Memory store	PostgreSQL + pgvector
Object storage	Amazon S3 or equivalent
Graph	Optional Neo4j / graph layer in later phase
Tool integration	MCP
Observability	OpenTelemetry + application logs/dashboard
Authentication	Enterprise identity provider / OAuth
Deployment	Docker + AWS or approved enterprise platform

20. Innovation & Differentiation

- Decision-aware memory instead of document-only RAG.
- Shared organizational memory across multiple agents.
- Context planning instead of simple top-k retrieval.
- Temporal awareness: current, historical and superseded knowledge.
- Source authority and evidence-backed answers.
- Contradiction detection across enterprise systems.
- Human-in-the-loop based on action risk.
- Agent registry, permissions and tool governance.
- Memory updates from outcomes and lessons learned.
- Designed as a reusable enterprise agent platform rather than a single chatbot.

21. Success Metrics

Metric	How to Measure
Answer accuracy	Human-verified answers against a gold dataset
Evidence coverage	Percentage of factual claims with valid sources
Context relevance	Human/evaluator rating of retrieved context
Freshness	Age of context at retrieval time
Time saved	Manual investigation time vs. agent-assisted workflow
Action success	Successful tool execution rate
Decision reuse	Number of new decisions supported by previous knowledge
Risk detection	Known risks identified by the system
User satisfaction	Manager feedback / task completion survey

22. Risks & Mitigations

Risk	Impact	Mitigation
Hallucination	High	Evidence grounding, confidence, evaluation suite
Stale memory	High	Freshness metadata, TTL, source refresh
Conflicting sources	High	Authority ranking + contradiction detection
Unauthorized data access	Critical	RBAC/ABAC + source permissions
Unsafe agent action	Critical	Risk-based approvals and tool policies
Poor retrieval	Medium	Hybrid retrieval + reranking + evaluation
Over-engineering	Medium	Strict MVP scope and staged roadmap
Vendor lock-in	Medium	Abstract Context API and agent/tool interfaces

23. Roadmap

Phase	Capabilities
Phase 1 – Hackathon MVP	Manager Agent, project context, decisions, risks, evidence, 2–3 sources
Phase 2 – Organizational Memory	Episodic, semantic, procedural and experiential memory; knowledge graph
Phase 3 – Agentic Execution	MCP tools, agent registry, approvals, identity and governed actions
Phase 4 – Enterprise Platform	Cross-project agents, enterprise governance, observability, evaluation and self-improving memory

24. Future Vision

The long-term vision is to establish Enterprise Context Brain as a shared context and control plane for enterprise AI agents. Instead of every agent independently searching documents and learning isolated context, agents operate from a common organizational memory with governed access, evidence, freshness, policies and continuous learning.

The platform can eventually support project management, engineering operations, data engineering, incident management, customer operations, compliance and executive decision intelligence through specialized agents.

25. Conclusion

Enterprise Context Brain addresses a fundamental limitation of enterprise AI: agents can reason well, but their usefulness depends on whether they have the right organizational context at the right time and under the right permissions.

By combining organizational memory, context intelligence, multi-agent orchestration, MCP-based tools, governance, observability and human approval, the solution moves beyond a traditional RAG chatbot toward a reusable enterprise agent platform.

The hackathon MVP will prove the core hypothesis: a manager can ask a complex project question and receive an evidence-backed understanding of what happened, why it happened, what changed, what risks exist and what action should be considered—without manually searching multiple systems.

26. Research References

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